

Solution Requirements

Date	2 July 2026
Team ID	SWTID-2026-7147
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users (Bank Credit Officer and Financial Analyst) securely log in using username and password before accessing the system.	High
2	Authorization Levels	Loan Applicants can submit applications. Bank Credit Officers can review predictions. Financial Analysts can view reports and analytics.	High
3	External Interfaces	Connects with the Flask web application, XGBoost prediction model, IBM Cloud deployment, and applicant database.	High
4	Transaction Processing	Accept applicant details, process them through the ML model, and instantly generate a loan approval or rejection prediction.	High
5	Reporting	Generate prediction results, applicant history, approval/rejection reports, and model performance reports.	Medium

6	Business Rules	Loan approval depends on applicant information such as income, credit history, employment status, education, loan amount, and other features using the trained XGBoost model.	High
7	Compliance to Laws or Regulations	Protect applicant data and follow banking data privacy and security guidelines.	High
8	Other	Store prediction results and maintain application history for future reference.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Predict loan approval quickly after the applicant submits the form.	Prediction generated in less than 3 seconds
2	Scalability	Support increasing numbers of applicants without affecting performance.	Supports 1,000+ applications per day
3	Security & Data Privacy	Protect applicant information using secure communication and controlled access.	HTTPS , encrypted data transmission, secure login
4	Reliability & Availability	Ensure the system is available whenever users need it.	99% system availability
5	Usability & Accessibility	Provide an easy-to-use web interface for applicants and bank staff.	Responsive interface with clear navigation
6	Other (Maintainability)	Allow easy model updates, maintenance, and deployment on IBM Cloud.	Easy deployment and future model replacement